

LGM data assimilation : a systematic sensitivity analysis

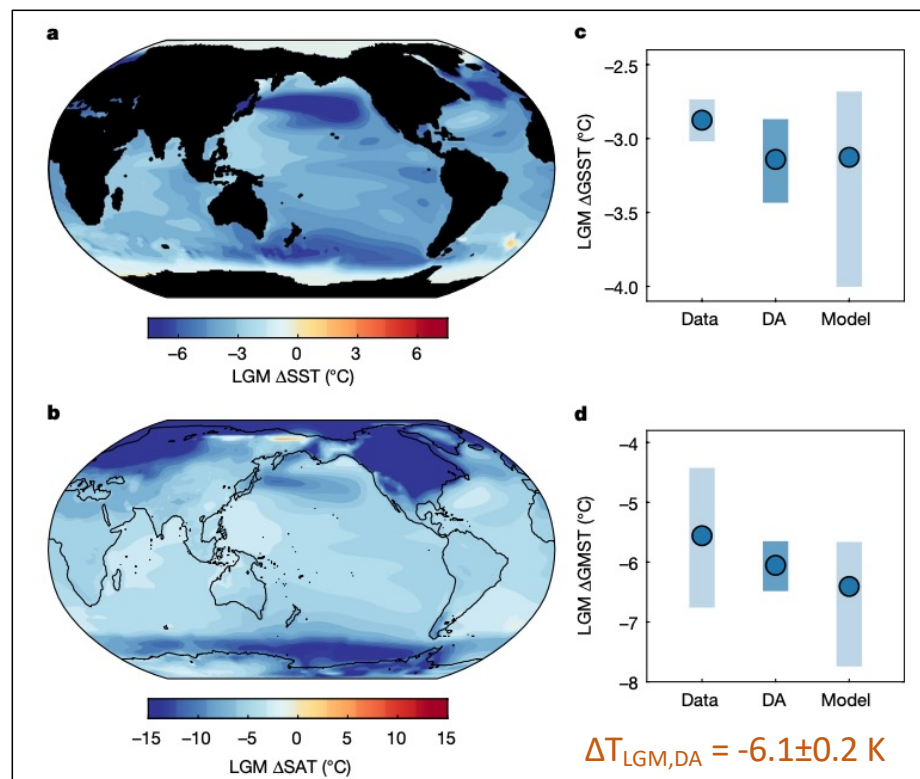
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Can DA really provide a robust estimate of LGM cooling?

- Sparse, uncertain proxy data
- Lack of diverse prior (not enough **good** model simulations)

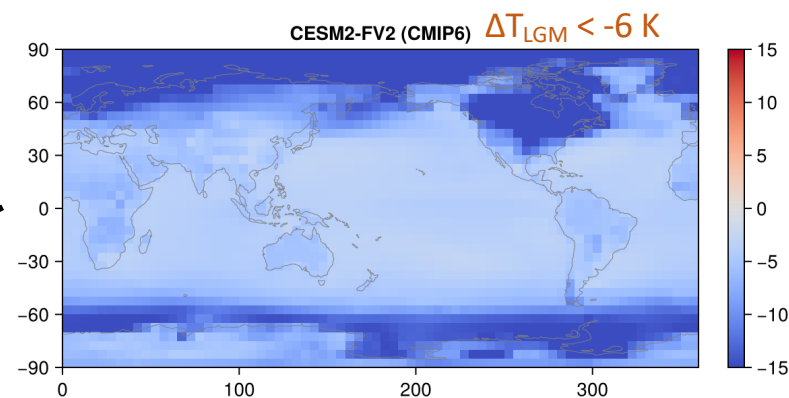
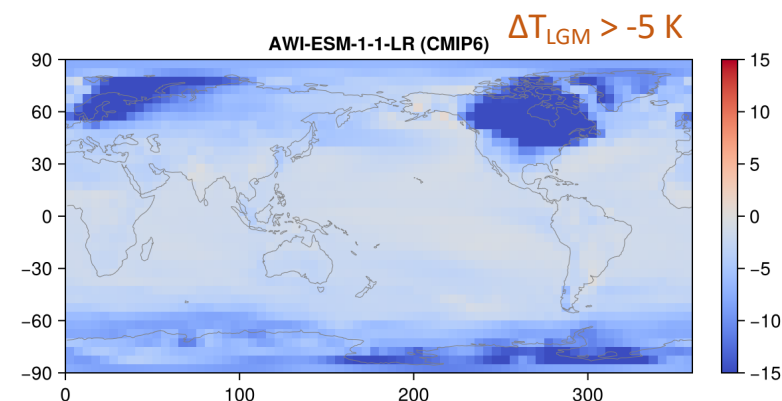
Goals:

- Understand key factors that influence reconstructions.
- Provide more robust estimates with traceable assumptions.
- Include new paleo proxy data – from terrestrial sources??

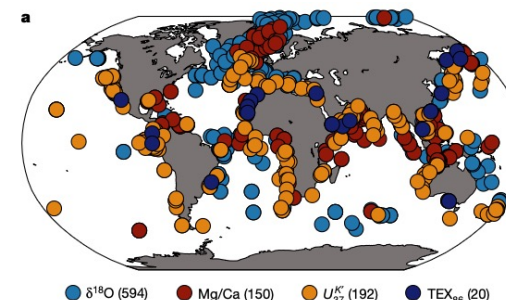


Tierney et al., 2020

Example ΔT_{LGM} fields from PMIP models
(6 models available from CMIP6)



Only used
CESM for
prior!



LGM marine
proxy sites

Tierney et al., 2020